



Specialization

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By Example: Implementing the Proposal

Limitations and Future Work

Implementation Details

Evaluation

Conclusions

# Employing Metaprogramming: A Macro-Driven Strategy to Overcome Compiler-Imposed Limitations on Specialization

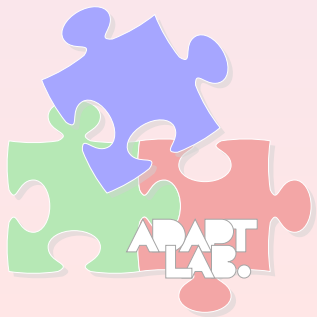
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# Specialization
## State of the Art and Limitations



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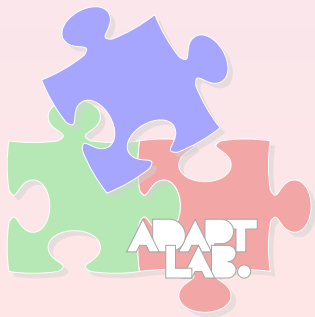
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Conclusions

Systems programming aims to combine generic programming with type-specific optimizations.

Mechanisms vary widely, from mature and widely used systems to experimental features, with varying degrees of success.

In many modern ecosystems, specialization is often unstable, forcing a tradeoff between performance and compiler stability.





# Problem Statement
## What's happening in Rust?



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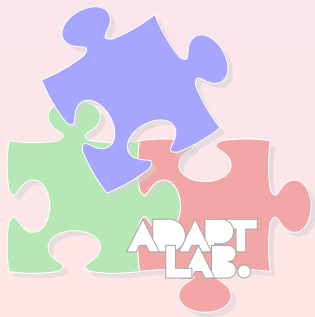
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In 2015, an RFC introduced specialization to Rust as an unstable feature confined to the Nightly Channel.

However, due to unresolved design issues and concerns about soundness of the system, specialization is still unstable.

Without stable specialization, developers often resort to code duplication or complex workarounds.







# Problem Statement
## Motivational Example

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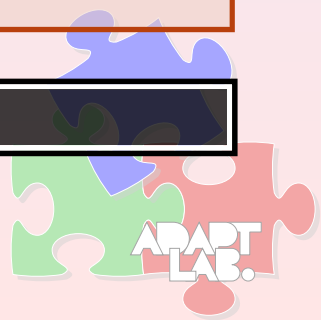
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Consider implementing a trait for different types, with a specialized implementation for a specific type.

```
1  trait MyTrait {
2      fn my_method(&self);
3  }
4  impl<T> MyTrait for T {
5      fn my_method(&self) {
6          println!("Generic implementation");
7      }
8  }
9  impl MyTrait for i32 {
10     fn my_method(&self) {
11         println!("Specialized for i32");
12     }
13 }
14 fn main() {
15     10.my_method();   // should print "Specialized for i32"
16     10.5.my_method(); // should print "Generic implementation"
17 }
```

```
error[E0119]: conflicting implementations of trait MyTrait for type i32
```





# Proposal
## Macro-Based Specialization



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Employ the powerful metaprogramming capabilities of Rust to build a macro library that simulates native specialization, with:

- Native-like look and feel
- Compile-time resolution
- Sound and stable
- Zero runtime overhead
- Powerful API

To do this, we use two macros: `#when` and `spec!`.

The approach can also be applied to other languages with powerful metaprogramming capabilities that lack native specialization support.





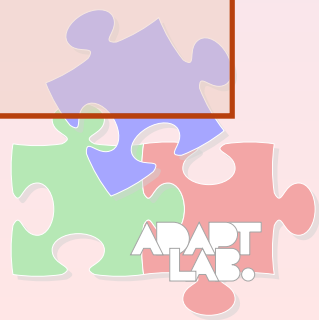


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# By Example: Implementing the Proposal
## Motivational Example

To illustrate how the proposed approach works in practice, let's revisit the motivational example we presented earlier.

```
1  trait MyTrait {
2      fn my_method(&self);
3  }
4  impl<T> MyTrait for T {
5      fn my_method(&self) {
6          println!("Generic implementation");
7      }
8  }
9  #[when(T = i32)]
10 impl<T> MyTrait for T {
11     fn my_method(&self) {
12         println!("Specialized for i32");
13     }
14 }
15 fn main() {
16     spec! { 10.my_method(); i32 };   // prints "Specialized for i32"
17     spec! { 10.5.my_method(); f64 }; // prints "Generic implementation"
18 }
```







# By Example: Implementing the Proposal

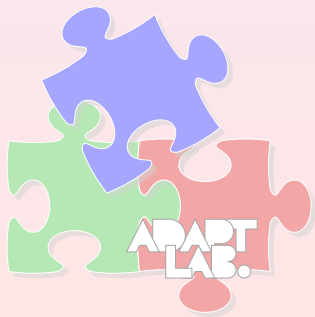
## Supported Predicates and Forms

The following predicates are supported:

- `T = Type`
- `all(p1, p2, ...)`
- `not(p)`
- `T: Tr1 + Tr2`
- `any(p1, p2, ...)`

We also support the following implementations forms:

- `impl<T> Trait for T`
- `impl<T> Trait for Type<T>`
- `impl<T> Trait<T> for Type`





# By Example: Implementing the Proposal
## Lifetimes



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# By Example: Implementing the Proposal



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# Limitations and Future Work



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# Implementation Details
## Conditions Definition



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# Implementation Details
## Method Wrapping



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# Implementation Details
## Build Scripts



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# Evaluation
## Approach



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# Evaluation
## Results



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# Conclusions



## Interesting results:

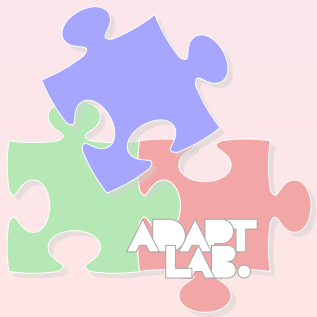
- We are writing an article (XXX) to be submitted to XXX.

## Interesting twist:

- XXX

## Future work:

- XXX







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# Thanks for your attention!

